

184. Inner Forward Error Correction (Inner FEC), type 800GBASE-LR1

184.1 Overview

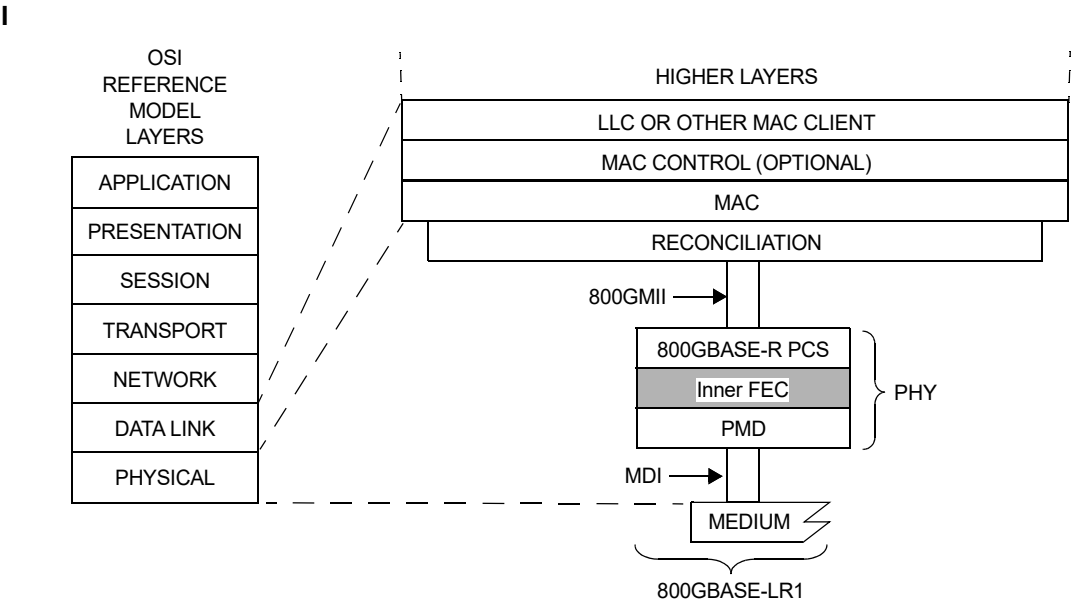
184.1.1 Scope

This clause specifies the Inner FEC sublayer for the Physical Layer implementations known as 800GBASE-LR1. This Inner FEC sublayer includes functionality often associated with the PMA sublayer.

When necessary to differentiate the Inner FEC defined in this clause from the 800GBASE-R Inner FEC defined in Clause 177, the term 800GBASE-LR1 Inner FEC is used. Within this clause the term Inner FEC refers specifically the 800GBASE-LR1 Inner FEC.

184.1.2 Position of the Inner FEC to the ISO OSI reference model

Figure 184–1 depicts the relationship of the Inner FEC sublayer (shown shaded), along with other Ethernet sublayers, to the ISO Open System Interconnection (OSI) reference model.



800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
FEC = FORWARD ERROR CORRECTION
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 184–1—Inner FEC relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

184.1.3 Summary of functions

The Inner FEC provides a second layer of FEC protection without decoding the data encoded by the RS-FEC in the PCS.

The Inner FEC provides the following services:

- Alignment lock, deskew, and reordering of 32 received PCS lanes
- Permutation of symbols across the 32 PCS lanes to create 32 Inner FEC flows
- Per flow convolutional interleaving (deinterleaving)
- Per flow BCH(126,110) encoding (decoding), and circular shift (inverse circular shift)
- DSP frame generation (transmit) and synchronization and termination (receive)
- Mapping (demapping) to (from) DP-16QAM symbols
- In the transmit direction, creates analog signals to drive the optical modulator in the PMD
- In the receive direction, processes the analog signals from the optical demodulator of the PMD

184.2 Inner FEC overview

A functional block diagram of the Inner FEC is shown in Figure 184–2.

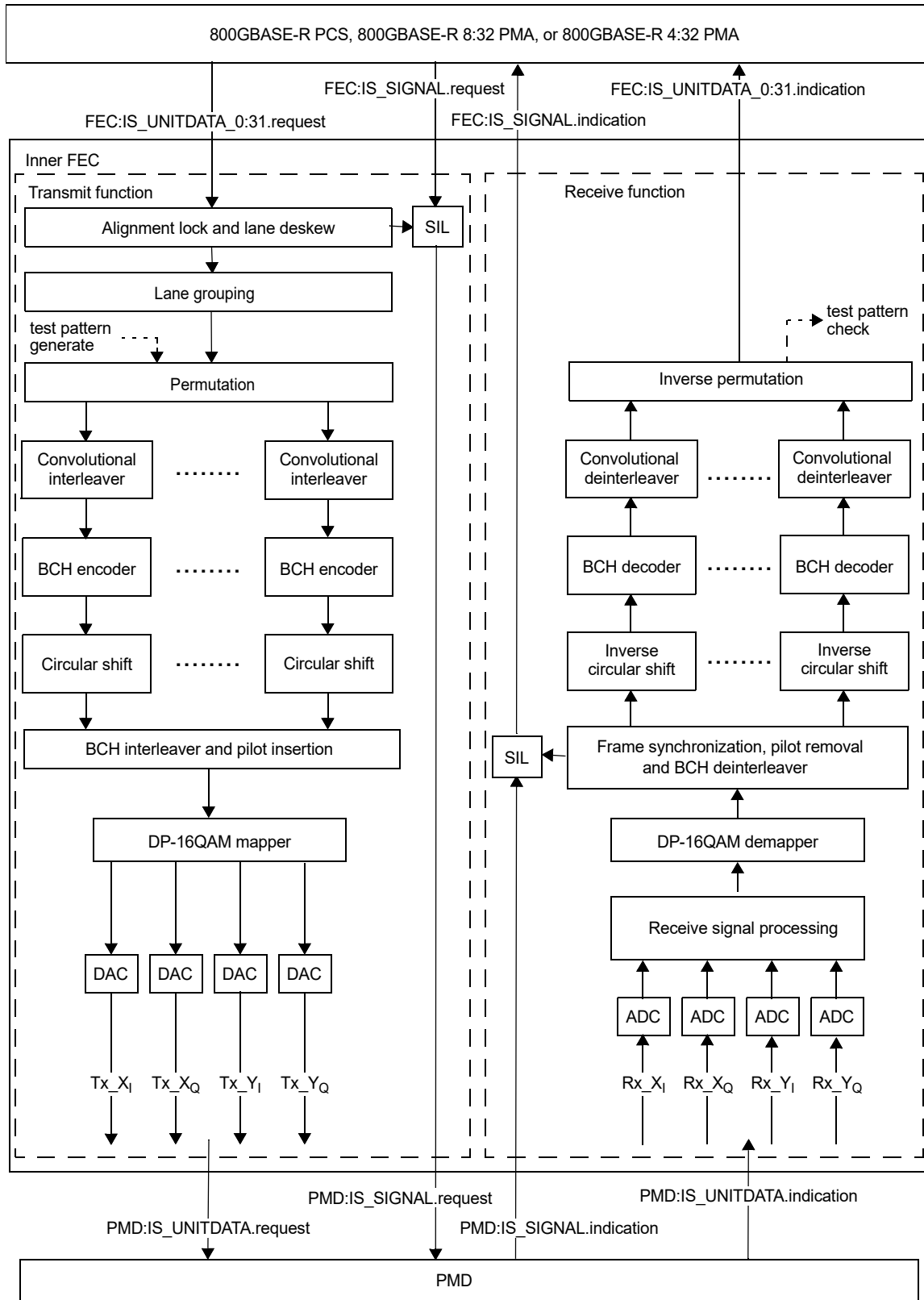


Figure 184-2—Inner FEC functional block diagram

The Inner FEC is composed of the Inner FEC transmit and the Inner FEC receive functions. Most of the encoding and decoding is done per Inner FEC flow.

The transmit functions receive 32 lanes of continuous RS(544,514) encoded 10-bit RS-FEC symbols. The lanes are reordered and deskewed. The RS-FEC symbols are then rearranged across the 32 lanes by a permutation function to create 32 Inner FEC flows. The RS-FEC symbols in each of the 32 flows are interleaved by a convolutional interleaver and then encoded with a BCH(126,110) FEC. A circular shift is performed before interleaving the encoded flows and generating the DSP frames for each polarization. The DSP frames are then mapped into 16QAM symbols, converted into four analog signals, and sent to the PMD via the PMD:IS_UNITDATA.request primitive.

In the receive direction, the Inner FEC synchronization process continually monitors the PMD:IS_SIGNAL.indication primitive. When the SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive is OK, it converts the four analog received signals into four-level signals and demaps the 16QAM symbols to recover the DSP frames. The synchronization process synchronizes on the received DSP frames, and recovers the information required to decode the bits mapped on each DP-16QAM symbol. It reverses the circular shift and BCH interleaving and performs decoding and error correction. Convolutional interleaving and permutation are undone to restore the original order of the lanes. The 32 lanes are forwarded to the client sublayer via the FEC:IS_UNITDATA.indication primitive along with the SIGNAL_OK parameter of the FEC:IS_SIGNAL.indication primitive.

184.3 Inner FEC service interface

The client sublayer above the Inner FEC in the Inner FEC functional block diagram (Figure 184–2) is a 800GBASE-R PCS, 800GBASE-R 8:32 PMA, or 800GBASE-R 4:32 PMA.

The service interface primitives are summarized as follows:

```
FEC:IS_UNITDATA_i.request(tx_symbol)
FEC:IS_UNITDATA_i.indication(rx_symbol)
FEC:IS_SIGNAL.request(SIGNAL_OK)
FEC:IS_SIGNAL.indication(SIGNAL_OK)
```

The primitives are defined for $i = 0$ to 31. In the transmit direction, the Inner FEC receives 32 parallel bit streams, each at a nominal signaling rate of 26.5625 Gb/s, from the client sublayer above. In the receive direction, the Inner FEC sends 32 parallel bit streams, each at a nominal signaling rate of 26.5625 Gb/s, to the client sublayer.

FEC:IS_UNITDATA_i.request is the same as PMA:IS_UNITDATA_i.request for the PMA 32:8 defined in 173.2.

FEC:IS_UNITDATA_i.indication is the same as PMA:IS_UNITDATA_i.indication for the PMA 32:8 defined in 173.2.

The Inner FEC sublayer receives signal status information from the client sublayer using the FEC:IS_SIGNAL.request primitive. The SIGNAL_OK parameter takes one of four values: OK, READY, IN_PROGRESS, or FAIL. When SIGNAL_OK is IN_PROGRESS or FAIL, the corresponding tx_symbol parameters on all lanes are unspecified.

The Inner FEC sublayer provides signal status information to the client sublayer using the FEC:IS_SIGNAL.indication primitive. The SIGNAL_OK parameter is set to one of four values by the

signal indication logic (SIL) according to Table 184–1. When the value of SIGNAL_OK is READY, IN_PROGRESS or FAIL, the corresponding rx_symbol parameters on all lanes are unspecified.

Table 184–1—FEC:IS_SIGNAL.indication(SIGNAL_OK) generation

PMD:IS_SIGNAL.indication SIGNAL_OK value	alignment_valid (see 184.8.2.1)	FEC:IS_SIGNAL.indication SIGNAL_OK
OK	true	OK
OK	false	READY
READY	don't care	READY
IN_PROGRESS	don't care	IN_PROGRESS
FAIL	don't care	FAIL

In addition to the primitives noted above, an associated clock is transferred from input to output along with the IS_UNITDATA primitives in the transmit and receive directions.

184.4 Service interface below the Inner FEC

The service interface below the Inner FEC is the PMD service interface for the PHY.

The PMD provides signal status information to the Inner FEC using the PMD:IS_SIGNAL.indication primitive. The SIGNAL_OK parameter takes one of four values: OK, READY, IN_PROGRESS, or FAIL. When SIGNAL_OK is IN_PROGRESS or FAIL, the rx_symbol parameters are undefined.

The Inner FEC sublayer provides signal status information to the PMD using the PMD:IS_SIGNAL.request primitive. The SIGNAL_OK parameter is set to one of four values by the SIL according to Table 184–2. When the value of SIGNAL_OK is READY, IN_PROGRESS or FAIL, the corresponding tx_symbol parameters on all lanes is unspecified.

Table 184–2—PMD:IS_SIGNAL.request(SIGNAL_OK) generation

FEC:IS_SIGNAL.request SIGNAL_OK value	all_locked (see 119.2.6.2.2)	PMD:IS_SIGNAL.request SIGNAL_OK
OK	true	OK
OK	false	READY
READY	don't care	READY
IN_PROGRESS	don't care	IN_PROGRESS
FAIL	don't care	FAIL

184.5 Transmit function

184.5.1 Alignment lock and deskew

The alignment lock and deskew function shall be identical to the processes specified in 172.2.5.1 except that only a deskew to 20-bit (two RS-FEC symbols) boundaries across all PCS lanes is required, and the codeword monitor function in Figure 172-6 is not required.

184.5.2 Lane grouping

The 32 Inner FEC sublayer interface bit streams might be received in any order. The bit stream number is defined by the unique portion (UM0 to UM5) of the alignment marker that is mapped to each PCS lane (see 172.2.4.6) carried by each bit stream.

The Inner FEC flows at the output of the lane grouping function, aligned to 10-bit RS-FEC symbols, are represented by $\text{pcsla}[q, i]$, where the index i represents the sequence of 10-bit RS-FEC symbols and q the Inner FEC flow number.

The lane grouping function shall organize the 32 bit streams into two groups according to the PCS lane number so that all bit streams with PCS lane numbers 0 to 15 are assigned to one group and all bit streams with PCS lane numbers 16 to 31 to another group. One group is then assigned to $\text{pcsla}[q, i]$ for $q = 0$ to 15 and the other is assigned to $\text{pcsla}[q, i]$ for $q = 16$ to 31.

184.5.3 Permutation

The lane permutation function shall rearrange the RS-FEC symbols of the $\text{pcsla}[q, i]$ Inner FEC flows to create 32 output flows such that each group of four consecutive symbols on each output flow contains one symbol from each of the four RS-FEC encoders in the 800GBASE-R PCS. Figure 184–3 shows an example of the lane permutation function operation.

The permutation function maps the RS-FEC symbols received on 32 pcsla flows to 32 output permo flows.

- Define $\text{permo}[q, i]$ to be the 10-bit symbol in output flow q at time i at the output of the permutation function.

The permutation function is defined by the following pseudocode:

```

For each  $i$ 
  For each  $q = 0$  to 31
     $\text{permo}[q, i] = \text{pcsla}[(q + 16 \times \text{floor}(i/2)) \bmod 32, i]$ 
  End for
End for

```

Where $\text{floor}(a)$ denotes the greatest integer less than or equal to the real number a .

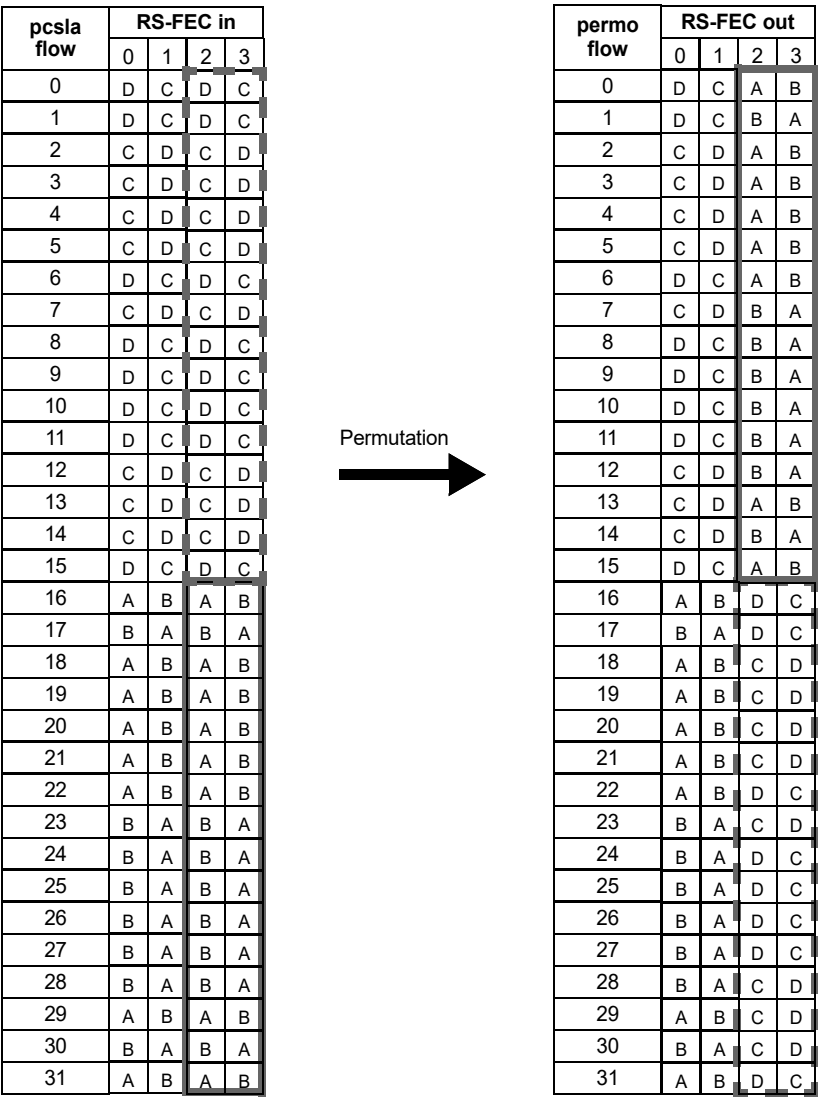


Figure 184–3—Example of the lane permutation function

184.5.4 Convolutional interleaver

The convolutional interleaver shall spread out the transmission order of consecutive units of 40 bits (four RS-FEC symbols) from the permo flows, such that the 110-bit input to the BCH encoder function contains no more than one RS-FEC symbol from any RS codeword. The convolutional interleaving function consists of three parallel delay lines (numbered 0 to 2), as illustrated in Figure 184–4. Each delay operator “D” represents a storage element of 40 bits. From one delay line to the next higher delay line, six delay operators are added. The bit ordering at the input of a delay line is preserved at the output.

The following list describes the convolutional interleaver process:

- The input and output switches are always aligned to the same row b , where $b = 0$ to 2
- A block of 40 bits is read from row b .
- The contents of row b are shifted to the right by 40 bits.
- A block of 40 bits is written to row b .
- The position of the switches is updated to $(b + 1) \bmod 3$

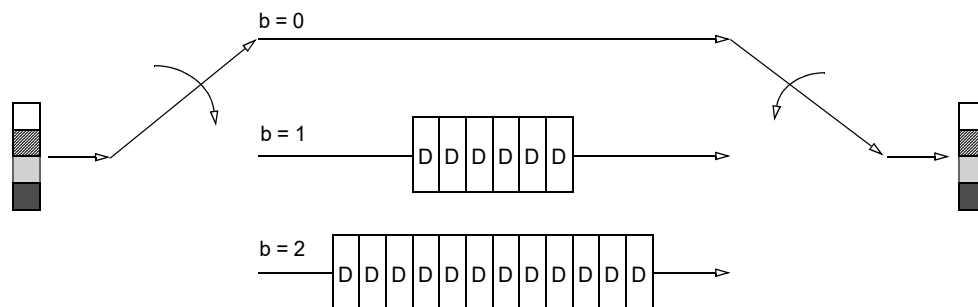


Figure 184-4—Convolutional interleaver

Initialization of the convolutional interleaver switches (to their topmost positions) is defined to occur at the start of every transmit DSP frame (see 184.5.8).

The following is performed individually on each of the q flows of permo, operating on 40-bit symbols (consisting of four RS-FEC symbols) j :

For each j
 $\text{convio}[j] = \text{permo}[j - 18 \times j \bmod 3]$
 End for

NOTE — $\text{convio}[j]$ is undefined when the index to permo is negative.

184.5.5 BCH encoder

The BCH encoder works in conjunction with the RS-FEC to provide a high performance FEC for 800GBASE-LR1.

The BCH code generator polynomial shall be derived starting with a BCH(127,113) code that is shortened to a BCH(124,110) code and then extended to the final BCH(126,110) code by adding even and odd parity check bits.

The code is defined as follows:

- Define c as a binary vector of length 126, and $c(x)$ a polynomial of degree 125 with the coefficients defined by c (where the bit 0 of c represents the coefficient of power 125)
- Then c is a codeword of the BCH(126,110) code if $c(x)$ is divisible (modulo 2) by the defined binary generator polynomial $g(x)$

BCH (126,110) code operates on 11 RS-FEC symbols provided by the convolutional interleaver

- Denote the index of the BCH codeword as u and the bit index within the BCH codeword as v where $v = 0$ to 125.

The encoding of each BCH codeword u on flow q ($q = 0$ to 31) of convio is defined as follows:

- A message polynomial $m(x)$ of degree 109 is defined where the coefficient of the x^{109-v} term is the BCH encoder convio[110 $\times$ u + v] for $v = 0$ to 109
- A generator polynomial $g(x)$ of degree 16 is defined as shown in Equation (184–1)
- A parity polynomial $p(x)$ of degree 15 shown in Equation (184–2), is defined as the remainder from the division (modulo 2) of $m(x) \times x^{16}$ by the generator polynomial shown in Equation (184–1).

$$g(x) = x^{16} + x^{14} + x^{11} + x^{10} + x^9 + x^7 + x^5 + x^3 + x + 1 \quad (184-1)$$

$$p(x) = p_{15} \times x^{15} + p_{14} \times x^{14} + \dots + p_1 \times x + p_0 \quad (184-2)$$

Define parity[15:0] to be the coefficients of the computed parity polynomial where parity[15] corresponds to p_{15} in Equation (184–2), parity[14] to p_{14} and so on.

The bit mapping between each convio input flow and encodeo output flow is:

```

For each  $u$ 
    encodeo[126 $u$ :126 $u$ +109] = convio[110 $u$ :110 $u$ +109]
    encodeo[126 $u$ +110:126 $u$ +125] = parity[15:0]
End for

```

184.5.6 Circular shift

The circular shift function shall be applied to each flow. It rearranges the 110 payload bits of each BCH FEC codeword to further increase robustness to burst errors. Consider each 126-bit BCH FEC codeword as a vector of bits, encodeo[j], and apply the following process:

```

For  $j = 0$  to 109
    circo[ $j$ ] = encodeo[( $j - 20q$ ) mod 110]
End for

For  $j = 110$  to 125
    circo[ $j$ ] = encodeo[ $j$ ]
End for

```

Where q corresponds to the flow number (0 to 31).

184.5.7 BCH interleaver

The BCH interleaver shall work in conjunction with the circular shift to improve robustness to burst errors and unequal bit error rates of the bits extracted from the DP-16QAM symbols, due to constellation position and different PDL noise on different polarizations.

The BCH interleaver spreads the bits of each BCH codeword evenly across all dimensions of the DP-16QAM symbols.

The BCH interleaver works on a block of 32 synchronized BCH codewords generated on the 32 circo flows.

- Denote the index of the 126-bit codeword block aligned across all 32 circo flows as k .
- Denote the index of the BCH interleaver output intero flows as r where $r = 0$ to 3.
- Denote $2d + j$ where $d = 0$ to 503 and $j = 0$ to 1, as an index of the bits within block k of output bits formed from input k block bits.

The mapping between each circo input flows and intero output flow is:

```

For each  $k$ 
  For  $r = 0$  to 3
    For  $d = 0$  to 503
      For  $j = 0$  to 1
        intero[ $r, 2d + j$ ] =
          circo[ $4 \times \text{floor}(d/63) + (2d + \text{floor}(d/2) \bmod 2 + r) \bmod 4, 2 \times d \bmod 63 + (d + j) \bmod 2$ ]
      End for
    End for
  End for
End for

```

184.5.8 DSP frame

The DSP frame is defined as a set of $96 \times 64 = 6144$ 4-bit blocks. 4-bit pilot symbols are inserted every 64 4-bit blocks (one 4-bit pilot symbol, 63 4-bit payload blocks). The details of the DSP frame are shown in Figure 184–5.

Two streams of DSP frames, one for each polarization, are generated by the Inner FEC. The 4-bit blocks for DSP frame_0 are constructed from two consecutive bits from intero[0] (to be mapped to X_I) and two consecutive bits from intero[1] (to be mapped to X_Q), and the 4-bit blocks for the DSP frame_1 are constructed from two consecutive bits from intero[2] (to be mapped to Y_I) and two consecutive bits from intero[3] (to be mapped to Y_Q). See 184.5.11.1 for the mapping function.

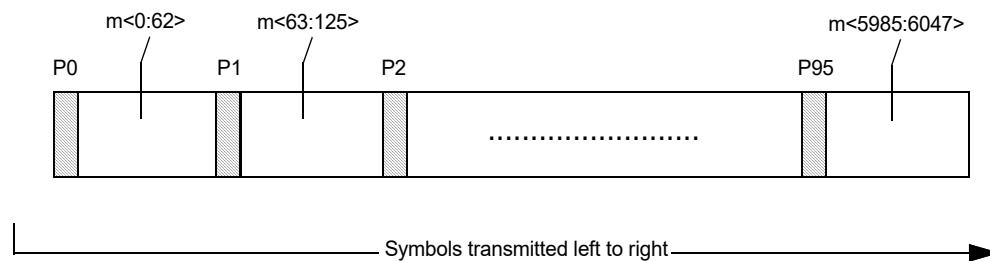


Figure 184–5—DSP frame

184.5.9 Pilot sequence

The pilot sequence shall be a fixed PRBS9 sequence with different seed values for DSP frame_0 and for DSP frame_1. The generator and seed values for each pilot sequence are shown in Figure 184–6 and listed in Table 184–3. The seeds are selected so that the pilot sequences are DC balanced.

For both DSP frame_0 and DSP frame_1, the generator is initialized using the seed at the start of every DSP frame. The generator produces a sequence of 192 bits. A zero bit is inserted after each bit to generate the bit-pairs that form the pilot symbols, which use the outer points of the 16QAM constellation. The bit-pairs that compose the pilot sequence are shown in Table 184–4. They are selected such that they produce symbols that use the outer 16QAM constellation points, as shown in Figure 184–7.

The bit-pairs for the X polarization are distributed in a round-robin manner to outputs 0 and 1. The bit-pairs for the Y polarization are distributed in a round-robin manner to outputs 2 and 3.

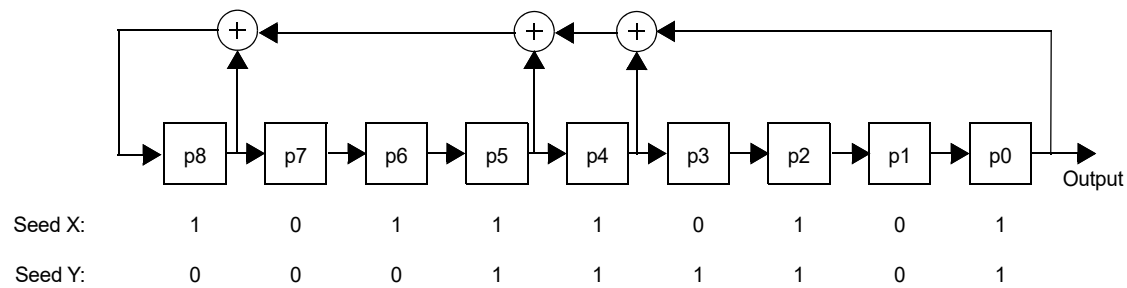


Figure 184-6—Pilot seed and sequence

Table 184-3—Pilot sequence generator polynomial and seed values

Generator polynomial	Seed for X-polarization	Seed for Y-polarization
$x^9 + x^8 + x^5 + x^4 + 1$	0x175	0x03D

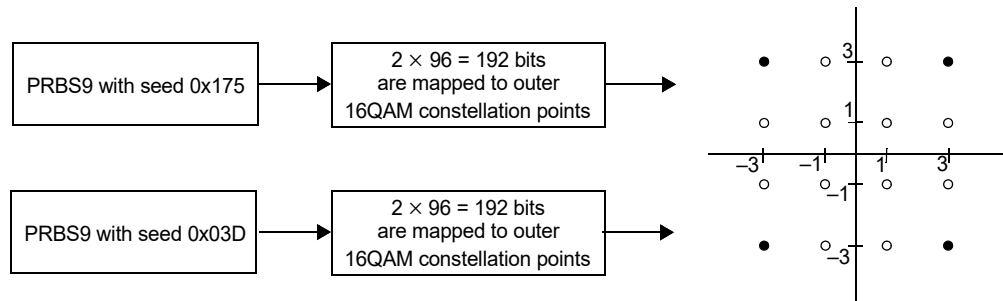


Figure 184-7—Pilot sequence mapping

Table 184-4—Pilot sequence

	Output					Output					Output					Output			
Index	0	1	2	3	Index	0	1	2	3	Index	0	1	2	3	Index	0	1	2	3
0	10	00	10	00	24	00	10	00	00	48	10	10	00	10	72	00	00	00	10
1	10	00	10	10	25	00	00	00	10	49	00	00	10	00	73	10	10	00	10
2	10	10	10	10	26	10	10	10	00	50	10	00	10	10	74	00	10	00	00
3	10	00	00	00	27	10	10	00	00	51	00	00	10	10	75	10	00	10	00
4	10	00	00	10	28	10	10	10	00	52	10	10	00	00	76	00	10	00	00

Table 184–4—Pilot sequence (*continued*)

Output					Output					Output					Output				
Index	0	1	2	3	Index	0	1	2	3	Index	0	1	2	3	Index	0	1	2	3
5	00	00	00	10	29	00	00	00	00	53	00	00	10	00	77	10	10	00	10
6	10	10	00	00	30	00	10	00	00	54	10	00	10	00	78	00	10	00	00
7	00	10	00	10	31	10	00	00	00	55	10	10	00	00	79	00	10	00	00
8	00	10	00	00	32	10	10	00	10	56	00	00	00	10	80	00	10	10	10
9	00	10	10	00	33	00	10	10	10	57	00	00	10	00	81	10	10	10	10
10	00	00	00	00	34	00	00	10	00	58	10	00	00	00	82	10	00	10	10
11	00	10	10	00	35	00	00	00	00	59	00	10	00	10	83	10	10	10	10
12	10	10	10	10	36	10	00	00	10	60	10	00	10	10	84	00	10	10	00
13	00	10	00	00	37	10	10	00	10	61	00	10	00	00	85	10	00	10	00
14	00	00	10	10	38	10	00	00	00	62	10	10	00	00	86	00	00	10	10
15	10	00	00	10	39	10	10	10	00	63	10	00	10	10	87	00	00	10	10
16	00	10	00	10	40	10	10	10	10	64	10	00	00	10	88	10	10	10	00
17	10	00	10	00	41	00	10	10	00	65	00	10	10	10	89	00	10	00	10
18	10	00	10	10	42	00	10	00	00	66	10	10	00	10	90	00	00	10	10
19	00	00	00	10	43	00	00	00	00	67	00	00	10	00	91	10	00	00	00
20	10	00	10	10	44	00	00	00	10	68	00	10	00	10	92	10	00	10	00
21	10	00	10	10	45	00	00	00	10	69	00	00	00	00	93	00	10	10	10
22	10	10	10	00	46	10	10	10	00	70	10	10	10	00	94	10	00	10	10
23	00	10	10	10	47	00	00	10	00	71	10	10	00	10	95	00	00	10	10

184.5.10 DSP frame generation

The BCH interleaver and pilot insert function shall map the bits received on intero flows to dspfo.

- Denote the index of the intero flows and the dspfo flows as q , where $q = 0$ to 3,
- Denote u as the index of the BCH codeword at intero and pilot inserter function aligned across the four intero flows, and aligned to the boundaries of the BCH encode function
- Denote the bit index within a DSP frame as v , where $v = 0$ to 127
- Denote the pilot sequence as $ps[q,k]$ for flow $dspfo[q]$ where $q = 0$ to 3, and pilot symbol index k where $k = 0$ to 95 (see Table 184–4)

The mapping between the ps and intero input flows, and the dspfo output flows is:

For every u

For $q = 0$ to 3

If $v = 0$, $dspfo[q, 128u + v] = ps[q, u \bmod 96]$

If $v = 1$, $dspfo[q, 128u + v] = 0$

For $v = 2$ to 127

dspfo[$q, 128u + v$] = intero[$q, 126u + v - 2$]
End for
End for
End for

184.5.11 DP-16QAM mapper

The DP-16QAM mapper shall convert every 2-bit pairs from the four dspfo flows to four flows of four-level signals. These four-level signals are used by the 800GBASE-LR1 PMD to generate a single optical DP-16QAM signal with orthogonal polarizations (see 185.5.2).

Table 184–5 shows the resulting 2-bit pair to four-level signal mapping.

Table 184–5—2-bit pair to four-level signal mapping

2-bit pair	Level
00	–3
01	–1
10	3
11	1

184.5.11.1 Symbol mapping to analog signals

The four signals from the DP-16QAM mapper shall be converted to four analog signals, two for each polarization: X_I , X_Q , Y_I , and Y_Q . The analog signals are passed to the 800GBASE-LR1 PMD sublayer using the tx_signal_xi, tx_signal_xq, tx_signal_yi, tx_signal_yq parameters of the PMD:IS_UNITDATA.request primitive.

184.6 Receive function

184.6.1 Receive signal sampling

The PMD:IS_UNITDATA.indication rx_signal_xi, rx_signal_xq, rx_signal_yi, and rx_signal_yq parameters are mapped to four analog signals Rx_X_I, Rx_X_Q, Rx_Y_I, and Rx_Y_Q.

Each Rx_X_I, Rx_X_Q, Rx_Y_I, and Rx_Y_Q signal shall be sampled at a sampling rate equal to or greater than the 800GBASE-LR1 signaling rate. The details of the sampling, including any quantization and the chosen sampling rate, are implementation specific.

The signals Rx_X_I, Rx_X_Q, Rx_Y_I, and Rx_Y_Q each carry a combination of the transmitting Inner FEC Tx_X_I, Tx_X_Q, Tx_Y_I, and Tx_Y_Q signals used by the transmitting PMD to generate the DP-16QAM symbols.

The resulting signals are forwarded to the receive signal processing function.

184.6.2 Receive signal processing

The four analog signals Rx_{X_I} , Rx_{X_Q} , Rx_{Y_I} , and Rx_{Y_Q} shall be processed and converted into two streams of 16QAM symbols representing the X and Y polarization of the DP-16QAM symbols.

The DP-16QAM symbols should be sampled with more than two bits per dimension, in order to enable the BCH decoder to correct errors and recover the bits from the symbols based on the mappings in Table 184–5.

Functions within the coherent receiver include, but are not limited to:

- Retiming and resampling to one sample per symbol after applying an antialiasing filter with a –3 dB bandwidth of 0.5 times the signaling rate to the incoming samples
- Equalization of signal distortion caused by chromatic dispersion, group delay, and polarization mode dispersion in the channel
- Clock phase recovery
- Frequency offset estimation and removal assuming a constant frequency offset over a given block size N which is implementation specific
- Carrier phase recovery
- Identification of the X and Y polarizations using the pilot sequences
- Identify the I and Q component of each polarization
- IQ offset evaluation and compensation

The details of the receive signal processing are implementation specific.

184.6.3 DP-16QAM demapper

The DP-16QAM demapper function shall map the four bit streams from the receive signal processing to 2-bit pairs along with any additional information required for BCH decoding. The precision of this information is implementation dependent.

The demapper implements the inverse of the DP-16QAM mapper function.

184.6.4 DSP frame synchronization and pilot removal

DSP synchronization shall comply with the DSP lock state diagram (Figure 184–9).

The pilot sequence bits are used to synchronize to the each of the two DSP frames, to identify the I and Q component of each polarization, to identify DSP frame_0 and DSP frame_1, and to remove the skew between polarizations. These symbols shall be removed prior to the BCH deinterleaver function.

The DSP frame synchronization process is restarted when synchronization is lost, the received SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive changes to FAIL, or when a reset of the Inner FEC is required (Figure 184–9).

184.6.5 BCH deinterleaver

The BCH deinterleaver shall perform the inverse of the BCH interleaver function (see 184.5.7) to map the information received on its four input flows to the correct location within its 32 output flows.

184.6.6 Inverse circular shift

The inverse circular shift shall perform the inverse of the circular shift function (see 184.5.6) to map the information received on its 32 input flows to the correct location within its 32 output flows. There are 32 inverse circular shift functions.

184.6.7 BCH decoder

The BCH decoder is a soft decision decoder that requires a higher resolution than two bits for each of the signals Rx_X_I, Rx_X_Q, Rx_Y_I, and Rx_Y_Q. The decoder evaluates the incoming codeword and determines the most likely message value.

The BCH decoder shall extract the information bits from the codeword, correct them as necessary, and discard the parity symbols. The BCH decoder correction capability is implementation specific. If the BCH decoder is not able to correct a codeword the parity bits are discarded and the information bits are passed on without correction.

There are 32 BCH decoder functions one for each flow.

The following counters shall be implemented to aid a network operator in determining the link quality.

Inner_FEC_corrected_cw_counter

A 32-bit counter that counts once for each corrected FEC codeword processed by any of the 32 BCH decoders when alignment_valid is true. A corrected Inner FEC codeword is a codeword that contained errors and was corrected or miscorrected. Note that when there is more than one bit error in a codeword there is a chance that the soft decision decoder could miscorrect the codeword. The Inner FEC decoder interprets miscorrected codewords as corrected codewords.

Inner_FEC_uncorrected_cw_counter

A 32-bit counter that counts once for each uncorrected FEC codeword processed by any of the 32 BCH decoders when alignment_valid is true. An uncorrected FEC codeword is a codeword that contains errors that the decoder was unable to correct.

Inner_FEC_total_bits_counter

A 64-bit counter that counts once for each bit processed by any of the 32 BCH decoders when alignment_valid is true. This may be used together with the FEC_corrected_bits_counter to calculate pre-FEC BER.

Inner_FEC_corrected_bits_counter

A 64-bit counter that counts once for each bit corrected by any of the 32 BCH decoders when alignment_valid is true. This may be used together with the FEC_total_bits_counter to calculate pre-FEC BER.

Inner_FEC_cw_counter

A 48-bit counter that counts once for each FEC codeword received when alignment_valid is true.

Inner_FEC_codeword_error_bin_k

A set of five 32-bit counters, where k = 0 to 4. When alignment_valid is true, for k = 0 to 3, Inner_FEC_codeword_error_bin_k counts once for each codeword received with exactly k bits detected in error. Inner_FEC_codeword_error_bin_4 increments when alignment_valid is true and four or more bits in an Inner FEC codeword are detected in error and are corrected. For example, if an Inner FEC codeword has exactly two bits detected in error, then Inner_FEC_codeword_error_bin_2 is incremented. Note that Inner_FEC_codeword_error_bin_0 is only incremented for codewords determined to be correct by the decoder (zero detected incorrect bits) and is not incremented for codewords determined to be uncorrectable which are counted in Inner_FEC_uncorrected_cw_counter.

For each codeword processed, exactly one counter in Inner_FEC_codeword_error_bin_k or Inner_FEC_uncorrected_cw_counter is incremented.

184.6.8 Convolutional deinterleaver

The convolutional deinterleaver shall perform the inverse of the convolutional interleaver function (see 184.5.4) to map the information received on its 32 input flows to the correct location within its 32 output flows. There are 32 convolutional deinterleaver functions.

The convolutional deinterleaving function consists of three parallel delay lines (numbered 0 to 2), as illustrated in Figure 184–8. Each delay operator “D” represents a storage element of 40 bits. From one delay line to the next higher delay line, six delay operators are added. The bit ordering at the input of a delay line is preserved at the output.

The following list describes the convolutional deinterleaver process:

- The input and output switches are always aligned to the same row b , where $b = 0$ to 2
- A block of 40 bits is read from row b .
- The contents of row b are shifted to the right by 40 bits.
- A block of 40 bits is written to row b .
- The position of the switches is updated to $(b + 1) \bmod 3$

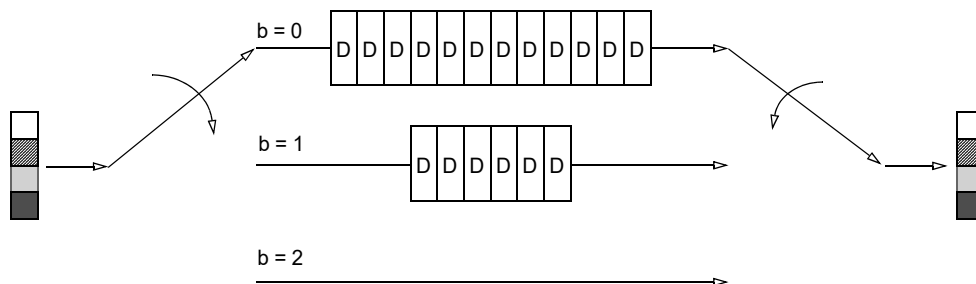


Figure 184–8—Convolutional deinterleaver

Initialization of the convolutional deinterleaver switches (to their topmost positions) is defined to occur at the start of every received DSP frame (see 184.5.8).

The following is performed individually on each of the 32 flows:

- Denote the input and output of the convolutional deinterleaver as $\text{input}[j]$ and $\text{output}[j]$, where the index j identifies 40-bit symbols.

For each j

$\text{output}[j + 18 \times (2 - j \bmod 3)] = \text{input}[j]$

End for

Output[j] is undefined when the index is negative.

184.6.9 Inverse permutation

The inverse permutation function shall map the bits received from the convolutional deinterleaver on input p , where $p = 0$ to 31, to 32 PCS lanes, and provide this data through the FEC:IS_UNITDATA_ i .indication primitives.

The mapping between the bit sequences on input p flows and the PCS lanes is the inverse of the permutation mapping function (see 184.5.3).

184.7 Test pattern

184.7.1 Test pattern generator

The Inner FEC shall include a PRBS31 test pattern generator (Figure 184–2). In test mode, 10-bit blocks from the PRBS31 generator are round-robin distributed to the 32 pcsla Inner FEC flows.

The PRBS31 generator implements the PRBS31 pattern defined by Equation (49–2) and shown in Figure 49–9.

The test pattern generator is enabled by the tx_generator_enable control variable.

184.7.2 Test pattern checker

The Inner FEC shall include a PRBS31 test pattern checker over the inverse permutation output PCS lanes (see 184.5.3). To recover the PRBS31 test pattern, 10-bit blocks are round-robin collected from the 32 inverse permutation function output flows.

The test pattern checker is enabled by the rx_checker_enable control variable.

The PRBS31 test pattern checker shall include block error detection and 17 related counters. Block error detection and behavior of the counters is defined in 174A.9.

The following counters shall be implemented:

Inner_FEC_test_block_error_bin_0_ k

A set of 16 48-bit counters where counter k counts once for each test block received with exactly k errored test symbols, $k = 0$ to 15.

Inner_FEC_test_block_error_bin_0_16p

A 48-bit counter that counts once for each test block received with 16 or more errored test symbols.

184.8 State diagrams

This subclause is composed of detailed state diagram processes, including the associated definition of constants, variables, functions and counters.

184.8.1 State diagram conventions

The notation used in the state diagrams follows the conventions of 21.5.

Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

184.8.2 State diagram definitions

184.8.2.1 Variables

alignment_valid

A Boolean variable that is set to true if the polarization symbol streams are aligned. Polarization symbol streams are considered to be aligned when $\text{dsp_lock}_{<x>}$ is true for both x , each polarization symbol stream has a unique identifier ($\text{dsp_ps_id}_{<0>}$ is different from $\text{dsp_ps_id}_{<1>}$), and the polarization symbol streams are deskewed. Otherwise, this variable is set to false.

current_ps_id

A variable that holds the polarization symbol stream identifier corresponding to the current pilot sequence that is recognized on a given polarization symbol stream.

Values:

0: Value of the pilot sequence for the given polarization symbol stream maps to Table 184–4 sequence X_I, X_Q .

1: Value of the pilot sequence for the given polarization symbol stream maps to Table 184–4 sequence Y_I, Y_Q .

dsp_ps_id $_{<x>}$

A variable that holds the value of the polarization symbol stream identifier received on the 800GBASE-LR1 PMD physical signal x , where $x = 0$ or $x = 1$. Its value is set by the DSP lock state diagram (Figure 184–9).

dsp_lock $_{<x>}$

A Boolean variable that indicates the receiver has detected the location of the pilot sequence for a given polarization symbol stream on the 800GBASE-LR1 PMD service interface, where $x = 0$ or 1. Its value is set by the DSP lock state diagram (Figure 184–9).

FEC_reset

A Boolean variable that is true when set by a management entity and is false otherwise.

reset

A Boolean variable that controls the resetting of the Inner FEC. It is true whenever a reset is necessary including when initiated by FEC_reset and during power on. Otherwise, this variable is set to false.

restart_lock

A Boolean variable that is used to restart the synchronization process in the DSP lock state diagram (Figure 184–9) for both polarization symbol streams when 8 pilot symbols in a row fail to match the expected values within either polarization symbol stream.

signal_ok

A Boolean variable that is set based on the most recently received SIGNAL_OK parameter of the $\text{PMD:IS_SIGNAL.indication}$ primitive. It is true if the value was OK and false if the value was FAIL.

sym_counter_done

A Boolean variable that is set to true when the counter sym_counter has reached its terminal count. It is set to false when the counter is started in the DSP lock state diagram (Figure 184–9).

sym_slip_done

A Boolean variable that indicates the next candidate pilot symbol position is available for testing. It is set to true when the SYM_SLIP function completes and is set to false upon entering the GET_SYMBOL state of the DSP lock state diagram (Figure 184–9).

sym_valid

A Boolean variable that is set to true if the received symbol is a valid pilot symbol according to the

state of the pilot sequences generator (see 184.5.9) for the value of the `current_ps_id` variable. Otherwise, this variable is set to false.

`test_ps`

A Boolean variable that is set to true when a candidate pilot symbol position is available for testing and false upon entering the `FIND_1ST` state of the DSP lock state diagram (Figure 184–9).

184.8.2.2 Functions

`FIRST_COMPARE`

This function initializes the pilot sequence generator (see 184.5.9) and compares the value of the first pilot symbol to the current symbol to determine if a pilot symbol has been detected and returns the result of the comparison using the variable `sym_valid`.

`POL_IDENTIFY`

This function compares the value of the second pilot symbol of each polarization sequence, to the current symbol to determine if a pilot symbol has been detected and returns the result of the comparison using the variable `sym_valid` and the identified polarization using the variable `current_ps_id`.

`SYM_COMPARE`

This function compares the value of the expected pilot symbol according to the state of the pilot sequence generator (see 184.5.9) and the value of the `current_ps_id` variable, to the current symbol to determine if a pilot symbol has been detected and returns the result of the comparison using the variable `sym_valid`.

`SYM_SLIP`

Causes the next candidate pilot symbol position to be tested. The precise method for determining the next candidate pilot symbol position is not specified and is implementation dependent. However, an implementation shall evaluate all possible symbol positions.

184.8.2.3 Counters

`ps_bad_count`

Counts the number of consecutive pilot symbols that don't match the expected value for a given polarization stream.

`ps_good_count`

Counts the number of consecutive pilot symbols that match the expected value for a given polarization stream.

`sym_counter`

This counter counts the interval of 64 symbols between the position of one pilot symbol and the expected position of the next pilot symbol on a polarization symbol stream. This counter starts with a value of zero when `start_sym_counter` is asserted, is incremented for each symbol on that polarization symbol stream, and reaches a terminal value of 64 at the expected position of the next pilot symbol on that polarization symbol stream.

184.8.3 State diagram figures

The Inner FEC shall implement two DSP frame lock processes as shown in Figure 184–9 to identify the boundaries of the DSP frames. The synchronization process operates independently on each polarization. The state diagram determines when the Inner FEC has detected the location of the pilot sequence in the signal received from the PMD service interface.

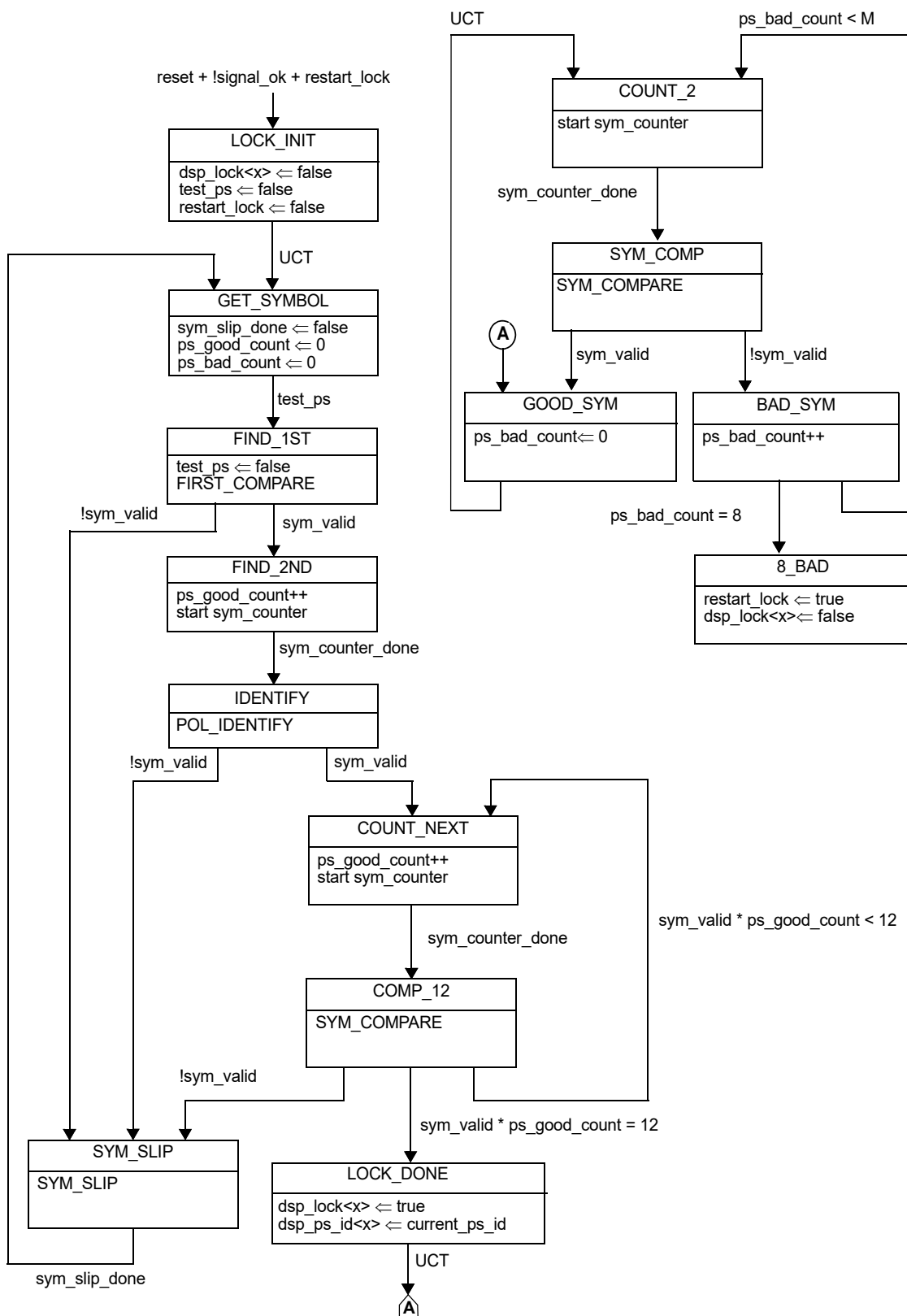


Figure 184-9—DSP lock state diagram

184.9 Delay constraints

The maximum delay contributed by the Inner FEC (sum of transmit and receive delays at one end of the link) shall be no more than 131 072 bit times (256 pause_quanta or 163.84 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references.

184.10 Path data delay for time synchronization

The Inner FEC sublayer optionally provides path data delay information for time synchronization (see Clause 90).

The ability to report Inner FEC path data delays is indicated by the Inner_FEC_delay_ns_TX_ability, Inner_FEC_delay_subns_TX_ability, Inner_FEC_delay_ns_RX_ability, and Inner_FEC_delay_subns_RX_ability status variables (see Table 184–7).

The Inner FEC sublayer has an intrinsic delay pattern where the sum of the transmit and receive delays is a constant value. As recommended in 90A.7, the maximum delay value is allocated to the transmit path and the minimum delay value to the receive path. The transmit and receive path data delay values are reported as the delay incurred by dspfo[3,1894] (see 184.5.10).

Four separate delays are reported in the following eight path data delay status variables (see Table 184–7):

- Inner_FEC_delay_ns_TX_max, Inner_FEC_delay_subns_TX_max
- Inner_FEC_delay_ns_TX_min, Inner_FEC_delay_subns_TX_min
- Inner_FEC_delay_ns_RX_max, Inner_FEC_delay_subns_RX_max
- Inner_FEC_delay_ns_RX_min, Inner_FEC_delay_subns_RX_min

184.11 Management variables

Inner FEC control and status variables intended to be accessible via a management system are listed in Table 184–6 and Table 184–7.

If the MDIO interface (see Clause 45) is implemented, Inner FEC control and status variables shall be mapped to the MDIO registers as shown in Table 184–6 and Table 184–7.

If the MDIO interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 184–6—Inner FEC control variables and MDIO mapping

Control variable	Variable reference	MDIO Register/bit number	MDIO register/bit reference
FEC_reset	184.8.2.1	1.0.15	45.2.1.1

Table 184–7—Inner FEC status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
dsp_lock<0>	184.8.2.1	1.2411.0	45.2.1.257
dsp_lock<1>	184.8.2.1	1.2411.1	45.2.1.257
alignment_valid	184.8.2.1	1.2410.0	45.2.1.256
tx_generator_enable	184.5.3	1.1501.3	45.2.1.170
rx_checker_enable	184.7.1	1.1501.0	45.2.1.170
Inner_FEC_corrected_cw_counter	184.6.7	1.2402, 1.2403	45.2.1.258
Inner_FEC_uncorrected_cw_counter	184.6.7	1.2404, 1.2405	45.2.1.259
Inner_FEC_total_bits_counter	184.6.7	1.2406, 1.2407, 1.2408, 1.2409	45.2.1.260
Inner_FEC_corrected_bits_counter	184.6.7	1.2410, 1.2411, 1.2412, 1.2413	45.2.1.260
Inner_FEC_cw_counter	184.6.7	1.2424, 1.2425, 1.2426	45.2.1.262
Inner_FEC_codeword_error_bin_0	184.6.7	1.2427, 1.2428,	45.2.1.263
Inner_FEC_codeword_error_bin_1	184.6.7	1.2429, 1.2430	45.2.1.263
Inner_FEC_codeword_error_bin_2	184.6.7	1.2431, 1.2432	45.2.1.263
Inner_FEC_codeword_error_bin_3	184.6.7	1.2433, 1.2434	45.2.1.263
Inner_FEC_codeword_error_bin_4	184.6.7	1.2435, 1.2436	45.2.1.263
Inner_FEC_test_block_error_bin_0_<0:15>	184.7.2	1.2644 to 1.2691	45.2.1.267
Inner_FEC_test_block_error_bin_0_16p	184.7.2	1.2692 to 1.2694	45.2.1.267
Inner_FEC_delay_ns_TX_max	184.10	1.1813, 1.1814	45.2.1.177a
Inner_FEC_delay_subns_TX_max	184.10	1.1817	45.2.1.177a
Inner_FEC_delay_ns_TX_min	184.10	1.1815, 1.1816	45.2.1.177a
Inner_FEC_delay_subns_TX_min	184.10	1.1818	45.2.1.177a

Table 184–7—Inner FEC status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Inner_FEC_delay_ns_RX_max	184.10	1.1819, 1.1820	45.2.1.177b
Inner_FEC_delay_subns_RX_max	184.10	1.1823	45.2.1.177b
Inner_FEC_delay_ns_RX_min	184.10	1.1821, 1.1822	45.2.1.177b
Inner_FEC_delay_subns_RX_min	184.10	1.1824	45.2.1.177b
Inner_FEC_delay_ns_RX_ability	184.10	1.1800.4	45.2.1.175
Inner_FEC_delay_subns_RX_ability	184.10	1.1800.5	45.2.1.175
Inner_FEC_delay_ns_TX_ability	184.10	1.1800.6	45.2.1.175
Inner_FEC_delay_subns_TX_ability	184.10	1.1800.7	45.2.1.175

184.12 Protocol implementation conformance statement (PICS) proforma for Clause 184, Inner FEC type 800GBASE-LR1¹⁷

184.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 184, Inner FEC sublayer type 800GBASE-LR1, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

184.12.2 Identification

184.12.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

184.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 184, Inner FEC sublayer type 800GBASE-LR1
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
-------------------	--

¹⁷*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

184.12.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*MD	MDIO	Clause 45	Registers and interface supported	O	Yes [] No []
DC	Delay constraints	184.9	Device conforms to delay constraints	M	Yes []

184.12.4 PICS proforma tables for Inner FEC Sublayer type 800GBASE-LR1

184.12.4.1 Transmit function

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	Align lock and deskew	184.5.1		M	Yes []
TF2	Lane grouping	184.5.2		M	Yes []
TF3	Test pattern generator	184.7.1		M	Yes []
TF4	Permutation	184.5.3		M	Yes []
TF5	Convolutional interleaver	184.5.4		M	Yes []
TF6	BCH encoder	184.5.5		M	Yes []
TF7	Circular shift	184.5.6		M	Yes []
TF8	BCH interleaver	184.5.7		M	Yes []
TF9	Pilot sequence	184.5.9		M	Yes []
TF10	DSP frame generation	184.5.10		M	Yes []
TF11	DP-16QAM mapper	184.5.11		M	Yes []
TF12	Symbol mapping to analog signals	184.5.11.1		M	Yes []

184.12.4.2 Receive function

Item	Feature	Subclause	Value/Comment	Status	Support
RF1	Receive signal sampling	184.6.1		M	Yes []
RF2	Receiver signal processing	184.6.2		M	Yes []
RF3	DP-16QAM demapper	184.6.3		M	Yes []
RF4	DSP frame synchronization	184.6.4		M	Yes []
RF5	Pilot symbols removal	184.6.4		M	Yes []
RF6	BCH deinterleaver	184.6.5		M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
RF7	Inverse circular shift	184.6.6		M	Yes []
RF8	BCH decoder	184.6.7		M	Yes []
RF9	Convolutional deinterleaver	184.6.8		M	Yes []
RF10	Inverse permutation	184.6.9		M	Yes []
RF11	Test pattern checker	184.7.2		M	Yes []

184.12.4.3 State diagrams

Item	Feature	Subclause	Value/Comment	Status	Support
SM1	DSP frame lock process	184.8.3	Implements two DSP frame lock processes as depicted in Figure 184–9	M	Yes []

184.12.4.4 Time synchronization

Item	Feature	Subclause	Value/Comment	Status	Support
*TS	Support for time synchronization	184.10		O	Yes [] No []
TS1	Path data delays	184.10	The transmit and receive path data delay values are reported as if the data delay measurement point (DDMP) occurs on dspfo[3,1894]	TS:M	Yes [] N/A []
TS2	Nanosecond resolution	184.10	Path data delays reported with nanosecond or better resolution.	TS:M	Yes [] N/A []
TS3	Sub-nanosecond resolution	184.10	Path data delays reported with sub-nanosecond resolution.	TS:O	Yes [] No [] N/A []

184.12.4.5 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	184.11		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	184.11	Table 184–6 and Table 184–7	MD:M	Yes [] N/A []